

Hemal Arora

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EDUCATION

Stanford University

B.S. Electrical Engineering (Hardware and Software Track) | GPA: 3.85/4.0

Stanford, CA

Expected June 2027

- Coursework: Robot Dynamics, Manipulation & Control (CS223A/CS225A/AA174A); Robot Perception & Computer Vision (CS227A/CS231N); Reinforcement Learning (CS238/CS234); Machine Learning & Statistical Inference (CS229/EE278); Computer Systems & Digital Design (CS107E/EE108/EE180/EE271)

EXPERIENCE

Mosaic Intelligence Labs, Inc. (Stealth Medical Robotics Startup)

Member of Technical Staff: Simulation & Perception

San Francisco, CA

Jun 2026 - Sep 2026

- Built physics-based sensor simulators and synthetic data pipelines for an endovascular robotic navigation system using NVIDIA OptiX; calibrated simulated X-ray and intravascular ultrasound imaging with clinical data; trained vision models on ~100K synthetic frames, achieving sim-to-real transfer during deployment in an animal study.
- Extended and scaled NVIDIA Newton/Warp XPBD Cosserat-rod simulations for realistic guidewire/catheter dynamics, implementing batched rod solving, coaxial instrument coupling, and debugging solver stability; worked with the Isaac for Healthcare team and scoped requirements for scaling large model-predictive planning rollouts.
- Built a real-time coupled instrument dynamics and imaging simulator, enabling hardware-in-the-loop validation of factor-graph-based localization and state estimation algorithms under partial observations.

Stanford Interactive Perception and Robot Learning Lab (IPRL)

Undergraduate Researcher: Advised by Prof. Jeannette Bohg

Stanford, CA

Mar 2026 - Present

- Co-developed ClothAtlas (submitted to ICRA 2027), a graph-conditioned flow-matching model for 3D deformable state estimation that reconstructs complete garment geometry from partial observations, preserves correspondences through occlusion, and generalizes to unseen mesh topologies.
- Transferred ClothAtlas directly from simulation to real-world garment manipulation without fine-tuning, localizing 85.1% of occluded grasp targets within 10 cm vs. 36.2% for dense descriptor matching.
- Researching failure-aware robot learning methods that retrieve relevant prior interactions from memory and use action-conditioned world models to guide action selection in VLA policies.

Multiply Labs

Robotics Software Engineer

San Francisco, CA

Jun 2025 - Dec 2025

- Developed a motion generation system for Multiply Labs' robotic cell therapy manufacturing cluster in NVIDIA Isaac Sim, using cuMotion to execute collision-free trajectories without hard-coded waypoints.
- Trained Diffusion Policy and Action Chunking Transformer (ACT) models on ~150 teleoperated demonstrations for precise cartridge insertion into millimeter-scale slots, deployed and evaluated on a UR10e robot.
- Developed a production-ready camera SDK with strict mypy typing and 100% test coverage, used for cluster-wide ArUco auto-calibration and collecting Unitree G1 datasets to fine-tune the NVIDIA GR00T VLA.

SELECTED PROJECTS

- **Autonomous Pickleball-Playing Robot [Video]:** Built a Franka FR3 + TidyBot++ mobile manipulator that tracks incoming balls, predicts intercepts, repositions its base, and replans velocity-controlled strikes at 100 Hz using damped least-squares IK and velocity-constrained Hermite trajectories.
- **Maximum Likelihood RL for Continuous Control Robotics [Report]:** Extended MaxRL to sparse-reward 7-DOF manipulation and benchmarked it against RLOO, combining goal-conditioned behavior cloning, hindsight experience replay, and policy-gradient fine-tuning.

Skills: Python, C++, C, PyTorch, ROS 2, Isaac Sim, MuJoCo, NVIDIA Warp, Newton, OptiX, SystemVerilog, KiCad